HARD · PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

05

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS

The table summarizes the distribution of age and assigned group for 90 participants in a study.

	0–9 **years**	10–19 **years**	20 + **years**	**Total**
Group A	5	17	8	30
Group B	6	8	16	30
Group C	19	5	6	30
Total	30	30	30	90

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

A. $\frac{5}{18}$

B. $\frac{5}{12}$

C. $\frac{17}{30}$

D. $\frac{5}{6}$

Q2 PROBLEM-SOLVING AND DATA ANALYSIS

The positive number a is 2,241% of the sum of the positive numbers b and c, and b is 83% of c. What percent of b is a?

A. 23.24%

B. 49.41%

C. 2,324%

D. 4,941%

Q3

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

Data set A consists of the heights of 75 objects and has a mean of 25 meters. Data set B consists of the heights of 50 objects and has a mean of 65 meters. Data set C consists of the heights of the 125 objects from data sets A and B. What is the mean, in meters, of data set C?

STUDENT-PRODUCED RESPONSE

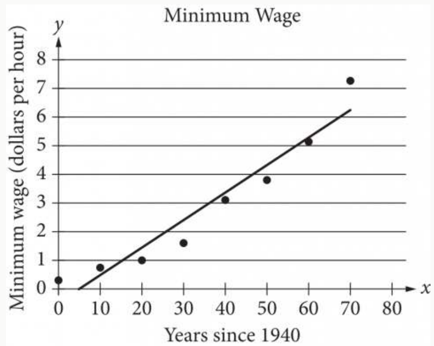
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

For an electric field passing through a flat surface perpendicular to it, the electric flux of the electric field through the surface is the product of the electric field's strength and the area of the surface. A certain flat surface consists of two adjacent squares, where the side length, in meters, of the larger square is 3 times the side length, in meters, of the smaller square. An electric field with strength 29.00 volts per meter passes uniformly through this surface, which is perpendicular to the electric field. If the total electric flux of the electric field through this surface is $4{,}640$ $\text{volts} \cdot \text{meters}$, what is the electric flux, in $\text{volts} \cdot \text{meters}$, of the electric field through the larger square?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



The scatterplot above shows the federal-mandated minimum wage every 10 years between 1940 and 2010. A line of best fit is shown, and its equation is $y = 0.096x - 0.488$. What does the line of best fit predict about the increase in the minimum wage over the 70-year period?

A. Each year between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

B. Each year between 1940 and 2010, the average increase in minimum wage was 0.49 dollars.

C. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

D. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.488 dollars.

Q6

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

On May 10, 2015, there were 83 million Internet subscribers in Nigeria. The major Internet providers were MTN, Globacom, Airtel, Etisalat, and Visafone. By September 30, 2015, the number of Internet subscribers in Nigeria had increased to 97 million. If an Internet subscriber in Nigeria on September 30, 2015, is selected at random, the probability that the person selected was an MTN subscriber is 0.43. There were p million MTN subscribers in Nigeria on September 30, 2015. To the nearest integer, what is the value of p ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

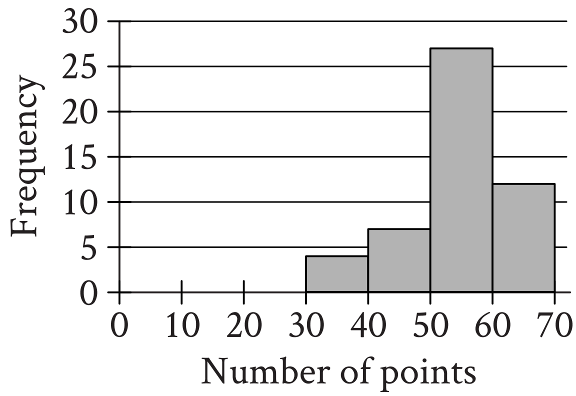
The value of a collectible comic book increased by 167% from the end of 2011 to the end of 2012 and then decreased by 16% from the end of 2012 to the end of 2013. What was the net percentage increase in the value of the collectible comic book from the end of 2011 to the end of 2013?

A. 124.28%

B. 140.28%

C. 151.00%

D. 209.72%



- The histogram has 4 bins.
- The frequency values of the 4 bins are as follows:
  - 30 to 39 points: More than halfway between 0 and 5
  - 40 to 49 points: About halfway between 5 and 10
  - 50 to 59 points: About halfway between 25 and 30
  - 60 to 69 points: About halfway between 10 and 15

The histogram summarizes data set A, which represents the number of points per player earned by 50 players of a game. A new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. Which of the following must be true?

I. The median number of points per player for data set B is less than the median number of points per player for data set A.
II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.

A. I only

B. II only

C. I and II

D. Neither I nor II

Q9

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

If $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$, what is the value of n?

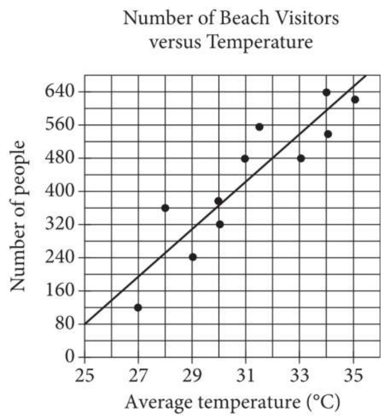
STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS



Each dot in the scatterplot above represents the temperature and the number of people who visited a beach in Lagos, Nigeria, on one of eleven different days. The line of best fit for the data is also shown. The line of best fit for the data has a slope of approximately 57. According to this estimate, how many additional people per day are predicted to visit the beach for each 5°C increase in average temperature?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.